EXPERIMENT:17-Experiment to understand Linear Regression for a given data set.

DESCRIPTION:

The program demonstrates Linear Regression by using study hours to predict marks. The dataset is divided into training and testing data, and a Linear Regression model is trained to make predictions.

PROGRAM:

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
# New dataset
data = {
    "StudyHours": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    "Marks": [35, 40, 45, 50, 55, 62, 68, 75, 82, 90]
}
df = pd.DataFrame(data)
print("Dataset:")
print(df)
# Dataset information
print("\nDataset Information:")
df.info()
# Statistical summary
print("\nSummary:")
print(df.describe())
# Features and label
features = df[["StudyHours"]].values
label = df["Marks"].values
# Split the data
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=42
)
```

```
# Create Linear Regression model
model = LinearRegression()
# Train the model
model.fit(x_train, y_train)
# Check model accuracy
print("\nTraining Score:", model.score(x_train, y_train))
print("Testing Score:", model.score(x_test, y_test))
# Coefficient and intercept
print("\nCoefficient:", model.coef_)
print("Intercept:", model.intercept_)
# Predict marks for new study hours
hours = float(input("\nEnter Study Hours: "))
prediction = model.predict([[hours]])
print("Predicted Marks:", prediction[0])
```

OUTPUT:

```
Dataset:
   StudyHours  Marks
0           1     35
1           2     40
2           3     45
3           4     50
4           5     55
5           6     62
6           7     68
7           8     75
8           9     82
9          10     90

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   StudyHours  10 non-null     int64
 1   Marks       10 non-null     int64
dtypes: int64(2)
memory usage: 232.0 bytes

Summary:
       StudyHours      Marks
count    10.00000  10.000000
mean      5.50000  60.200000
std       3.02765  18.413763
min       1.00000  35.000000
25%       3.25000  46.250000
50%       5.50000  58.500000
75%       7.75000  73.250000
max      10.00000  90.000000

Training Score: 0.9909737243924164
Testing Score: 0.9975259921646027

Coefficient: [6.0862069]
Intercept: 26.525862068965516

Enter Study Hours:  5
Predicted Marks: 56.956896551724135
```